Entity Relationship Diagram

Data dictionary and relationship specification for the QuoteIt application

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1. Entity relationship diagram

QuoteIt stores its domain data in three related tables. A user authors many quotes, and a user can like many quotes. The quote_likes table resolves that many-to-many relationship and records when each like was created.

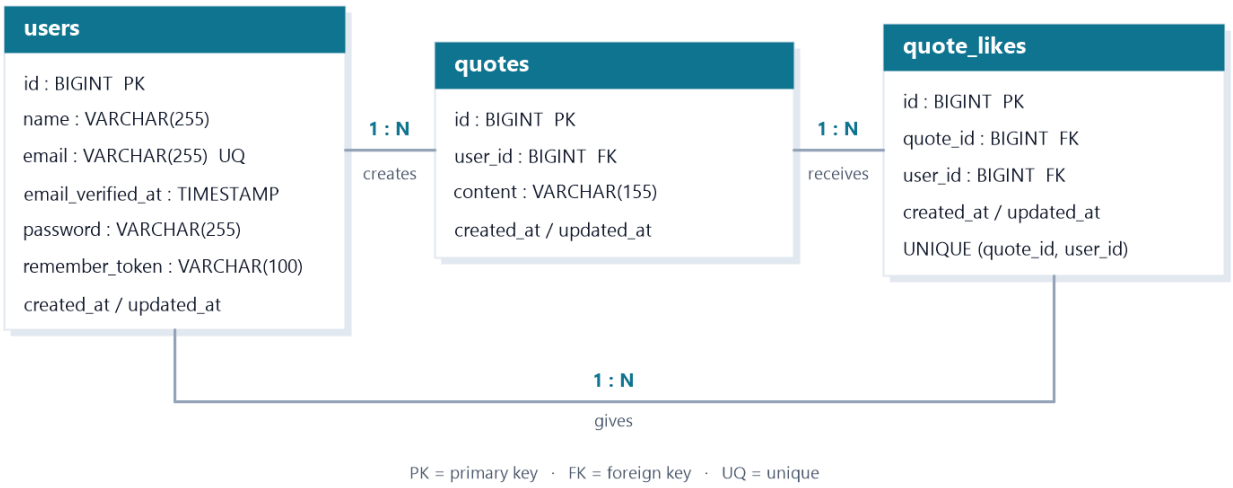


Figure 1. Entity relationship diagram for the QuoteIt domain model.

2. Relationship summary

Relationship	Cardinality	Key / table	On delete	Description
Users -> Quotes	1 : N	quotes.user_id -> users.id	CASCADE	A user authors zero or more quotes; each quote has exactly one author.
Users -> Quote likes	1 : N	quote_likes.user_id -> users.id	CASCADE	A user casts zero or more likes; each like belongs to one user.
Quotes -> Quote likes	1 : N	quote_likes.quote_id -> quotes.id	CASCADE	A quote receives zero or more likes; each like targets one quote.
Users <-> Quotes	M : N	quote_likes(quote_id, user_id)	CASCADE	Resolved by the quote_likes associative table; one row per user-quote pair.

quote_likes is an associative entity. Its composite unique index on (quote_id, user_id) guarantees that a user can like a given quote at most once, while any number of users may like the same quote.

PK = primary key | FK = foreign key | UQ = unique

Cardinality: 1 : N means one parent row is referenced by many child rows. The M : N relationship between users and quotes is resolved by the quote_likes table.

3. Data dictionary

The table lists every application column, its SQL type, its key or constraint, and whether it accepts NULL. Laravel timestamp columns are nullable and are managed automatically by Eloquent.

Table	Column	Type	Key / constraint	Null	Description
users	id	BIGINT UNSIGNED	PK, auto-increment	No	Primary key.
users	name	VARCHAR(255)	-	No	Display name.
users	email	VARCHAR(255)	UNIQUE	No	Login identifier.
users	email_verified_at	TIMESTAMP	-	Yes	Verification time.
users	password	VARCHAR(255)	-	No	Hashed password.
users	remember_token	VARCHAR(100)	-	Yes	Remember-me token.
users	created_at	TIMESTAMP	-	Yes	Created timestamp.
users	updated_at	TIMESTAMP	-	Yes	Updated timestamp.
quotes	id	BIGINT UNSIGNED	PK, auto-increment	No	Primary key.
quotes	user_id	BIGINT UNSIGNED	FK -> users.id (CASCADE)	No	Author of the quote.
quotes	content	VARCHAR(155)	-	No	Quote text, max 155.
quotes	created_at	TIMESTAMP	-	Yes	Created timestamp.
quotes	updated_at	TIMESTAMP	-	Yes	Updated timestamp.
quote_likes	id	BIGINT UNSIGNED	PK, auto-increment	No	Primary key.
quote_likes	quote_id	BIGINT UNSIGNED	FK -> quotes.id (CASCADE)	No	Liked quote.
quote_likes	user_id	BIGINT UNSIGNED	FK -> users.id (CASCADE)	No	User who liked.
quote_likes	created_at	TIMESTAMP	-	Yes	Liked timestamp.
quote_likes	updated_at	TIMESTAMP	-	Yes	Updated timestamp.

4. Keys, indexes, and integrity rules

The migrations declare the following database-level constraints. Cascading deletes keep the dataset consistent when a user or quote is removed.

Object	Type	Definition	Purpose
users_email_unique	Unique index	users(email)	Prevents duplicate accounts.
quotes_user_id_foreign	Foreign key	quotes(user_id) -> users(id) ON DELETE CASCADE	Deletes a user's quotes with the account.
quote_likes_quote_id_foreign	Foreign key	quote_likes(quote_id) -> quotes(id) ON DELETE CASCADE	Deletes likes when a quote is deleted.
quote_likes_user_id_foreign	Foreign key	quote_likes(user_id) -> users(id) ON DELETE CASCADE	Deletes likes cast by a deleted user.
quote_likes_quote_id_user_id_unique	Unique index	quote_likes(quote_id, user_id)	One like per user per quote.

5. Design notes

- Primary keys use Laravel's $table->id(): unsigned BIGINT, auto-increment.
- created_at and updated_at are nullable Eloquent-managed timestamps.
- Models live in app/Models; quote_likes is the pivot for likes and likedQuotes.